Analyzing Visitor Patterns at Mensa Shedhalle from Public Network Activity

Daniel Bischoff [*1] **Sarthak Bhardwaj** [*2] **Mihai Falcusan** [*3] **Philipp Langsteiner** [*4]

Abstract

Is it possible to know how crowded the mensa is and how long the queues are? Is there a regular pattern of visitor and is it possible to answer this question based on network activity? For investigating this questions network data was tracked during a regular working week. We estimate the total number of people having lunch using public WiFi traffic and use it to model queue waiting times. Through this, we reveal the optimal times for visiting the canteen and the advantages and limitations of single-sensor tracking via open-source software.The current paper shows how tracking public WiFi is privately possible and how the resulting data is useful for producing an estimate for human traffic flow in public spaces. You can find our code here.

1. Introduction

In our rapidly evolving modern society, efficient management of public spaces is crucial for optimizing infrastructure, enhancing public safety, and promoting sustainable urban development. As students of the Tübingen University, we noticed our own opportunity to improve the experience on the university campus, related to our canteen, the Mensa Shedhalle. Compared to past semesters, a common impression was that we had to wait longer in queue before getting our food and these queues seemed to form at different, but equally inconvenient times on each day of the week.

The current observational study is therefore aimed at investigating the student's experience at the university canteen and uncover their visiting patterns. One common technique for modelling human interaction with public spaces is by using WiFi or Bluetooth data to measure traffic flow (Traunmueller et al., 2018; Sapiezynski et al., 2015). The advantages of such methods are that they can provide detailed spatio-temporal information about people, although most literature either uses multiple directional scanners for estimation (Weppner et al., 2016), or possesses admin rights for the Wi-Fi access points (Wang et al., 2019). As such techniques are difficult to implement, we will explore single sensor tracking of public network data to also uncover the possibilities and limitations of such methods.

Because of the university lecturing schedule, we assume the number of people will have a pattern specific for each day of the week, while we expect weeks to be mostly identically distributed. We also want to understand when queues form based on this estimated number of people and model this from a statistical perspective to find out the best visiting times at the Shedhalle.

2. Data and Methods

Tracking procedure

For estimating the count of people, we looked at ways to capture the number of smartphones at the Mensa. To control for holiday effects, the tracking was performed during the second week from January 2024 between the 8th to the 12th. It would start at 11:00h, before the average student goes to eat, and stop at 14:00h when the employees stop serving the food. For tracking public network data, we used the same WiFi stick connected to our personal laptops and sat in the same spot. To be able to validate our analysis later, we also visually kept track of queues and counted the number of seconds it would take for us to get food.
For the network data, we initially started with B.L.E. (Bluetooth low energy scanners) to count the number of people in the Mensa. Using Bluetooth proved to have multiple drawbacks, like difficulties in differentiating the devices, limited capturing range and major underestimation of number of people as not so many have their Bluetooth on. This led us to switch to WiFi for our data collection, which solved our problem related to range and device differentiation.
The nature of WiFi data led us to capture the network activity of all devices on the relevant WiFi network (eduraom, UTGuest, Guest). We collected this WiFi network data using Wireshark. This powerful open-source software can display and monitor the WiFi and other network activity of the device.

Data Processing & Analysis

Wireshark stores the collected data in its own unique data format. It can record all WiFi packages within a range, providing rich network information. To prepare the data for our analysis, we created a script that transforms the Wireshark data into a CSV file. We used Tshark for this task, which is a native tool from Wireshark that aids data conversion.

Time	Source	Destination	Protocol	SSID
11:05	mac A	mac B	802.11	eduroam
11:06	mac C	mac D	802.11	UTGuest
11:06		mac D	802.11	UTGuest

Table 1. CSV layout example of the network data with censored examples. The source ID in the last row is missing for the sake of saving data, as source mac C is sending more than one message to destination device D.

Table 1 shows the format of CSV layout that we used to perform the analysis on. Cleaning the collected data was an important step, as the time column was very detailed ("Jan 12, 2024 10:57:39.977602000 W. Europe Standard Time") and bringing it to a useful format was an essential step for computing unique MAC ID count.

	Mon	Tue	Wed	Thu	Fri
Size	1.36GB	1.12GB	1.38GB	4.21GB	2.76GB
Entries	13.87M	1.35M	15.34M	9.70M	2.57M
Uniques	527 K	161 K	326 K	104 K	343 K
Cleaned	1716	3565	1914	2047	2763

Table 2. Original file size, entries, number of unique source addresses and the number of unique source addresses after we cleaned the data of each day.

To identify the number of unique devices, we filtered for packages of a specific network. Packages that are associated with a network have a Service Set Identifier (SSID), which is equivalent to the name of the network. We used these SSIDs to identify packages from networks of interest. Filtering for a specific network is required in an area with multiple networks, since almost all modern mobile phones randomize their MAC ID (Martin et al., 2017).
According to Android (2024), most devices using their software broadcast a random MAC address to find nearby access points. If an access point with a known SSID broadcasts back, they connect to another random, but for that network persistent MAC address. According to Apple (2021) this is the algorithm that IOS devices use as well. This information makes around 83% (Statista, 2020) of all smartphone users identifiable, since we could not find any information on the randomization of Huawei devices. These devices have at least as many mac addresses as their known networks, and multiple random probing MAC addresses. The number of unfiltered unique source mac addresses in the recorded data therefore ranges from 100.000 to 500.000 for the recorded days. To put this into perspective, if we count each MAC address as one person, the Mensa Shedhalle would have been visited by 1.14 times the population of Tübingen (Tübingen, 2024) on the day with the least addresses.
Filtering for a specific network is therefore imperative for obtaining a representative count of people. To understand why filtering for a network gives a correct device count, one needs to understand how a device connects to a network. When a device first broadcast their random MAC address to ping any Access Point around them, they either do not provide an SSID or hide the SSID, which then shows up in our data as "<MISSING>". If they find a known network, they communicate back to the access point using the internally assigned MAC address as the source address. This last communication also shows the SSID of the network in the package. Filtering for that SSID allows us to count how many devices have connected to a specific network. The random probing MAC addresses are ignored, since when filtering for the desired SSID, the address shows up in the destination column. If the device uses the internally assigned address for the network, the MAC address will appear in the source column. We use this property to extract the number of devices that connected to a specific network, by filtering for the SSID of that network and counting the number of unique source addresses.
On the cleaned data, we started by splitting the day-wise data into first and last seen data-frames, to find the count of unique mac addresses seen per-minute. This gave us an estimate of new entries and exits per-min in the canteen's network. Following that, to obtain the total count of unique mac addresses at any given time, we computed the cumulative sum of entry and exit counts in each minute, and subtracted the sum of last seen from the first seen, which can be observed in Figure 2.
To estimate the number of people per unique MAC address

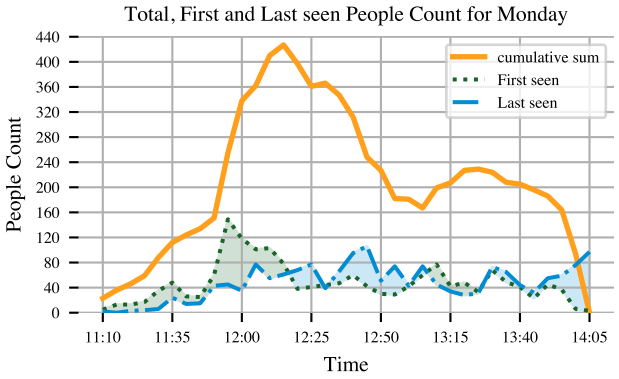


Figure 1. Illustration of estimated total, first and last seen people count. When the first count is higher than the last count, the total count increases and vice versa .

for each minute and quantify the uncertainty of our estimate, we decided to assign a probability of a unique MAC id representing one person, using a discrete probability distribution shown in Equation 1.

$$\xi \in [0,1] \quad , \quad f(\xi) = \begin{cases} 1, & \text{if } \xi < 0.83 \\ 2, & \text{if } 0.83 \leq \xi < 0.90 \\ 0, & \text{otherwise} \end{cases} \tag{1}$$

This distribution allows estimating the count of people based

on the number of unique MAC ids seen each minute. We based this distribution on 98% of students in Germany carrying a smartphone (Statista, 2020). Based on our observation and assumptions, around 10% of students either used their laptop or have and additional device. Additionally, we are assigning an additional 2% + 5% to account for people not having a phone or not enabling WiFi. This allowed us to get a better estimate of the number of people in the canteen via bootstrapping.

QUEUE MODEL

Our baseline model for the queue is described in equation 2.

$$\text{Queue}_t = \max(0, \sum_{t=0}^{n}(\text{ent}_{t-1} + \text{ent}_t) - \text{servspeed}_t) \quad (2)$$

In this baseline model, we assumed that the serving speed is fixed to a value of 6 servings per counter, as we observed in our real-world measurements. The queue in time point t is then computed as the number of people who did not get served in time point t-1. To this, the incoming number of people in time t are added, and the serving capacity of the canteen employees is subtracted. The purpose of this model was to see if we can obtain a first useful estimate for a queue. To improve this first model, we formulated a function allowing the serving speed to vary deterministically. It takes values between 8 and 20 servings per minute for all three counters together. This change in serving speed is dependent on the number of people entering the canteen. We observed that while a queue forms around peak times, there are 2 or 3 employees working at each counter with in advanced prepared servings at the start. We also included a serving speed upper bound at 20 servings per minute to model the physical cap on how fast people can be served due to the layout of the Mensa.

$$\begin{gathered}\text{servspeed}_t = \min(20, 8 + \sum_{t=0}^{n}(\text{ent}_{t-1} + \text{ent}_t)/3) \\ \text{with } 0 \leq \text{servspeed}_t \leq 20\end{gathered} \quad (3)$$

The next model tries to account for unknown factors causing variation in serving speed (e.g. someone's card stops working, someone asks for a bigger portion, a plate is already served, etc.) to produce better estimations. For this, we used a Gaussian distribution to model the random change in serving speed. For the mean of the Gaussian, we used the observed mean estimate of 5 servings over all counters. For the standard deviation, we looked at the variation in serving speed in each minute and fixed it to a value of 1 since there are three counters, which then results in an increase of 2 serves per counter using the three sigma rule. It would also account for the variations that cause the serving speed to decrease.

$$\begin{gathered}\text{servspeed}_t = \min(20, \text{srv}_t + \sum_{t=0}^{n}(\text{ent}_{t-1} + \text{ent}_t)/3) \\ \text{with } \text{srv}_t \in \mathcal{N}(6, 1) \text{ and } 0 \leq \text{servspeed}_t \leq 20\end{gathered} \quad (4)$$

For computing the standard error of this random serving speed parameter and for the overall count of people per MAC address, we used bootstrapping. Specifically, we ran 1000 simualtions to estimate the number of people in each second via 1 and used this estimate to compute queues. For the last model, we also computed the variation in serving speed in each second. Finally, we computed the mean estimated number of people in the queue and the standard error in each second together with 95% confidence intervals around these estimates as shown in (Fahrmeir et al., 2016).

3. Results

Figure 1 illustrates the computation of total people for Monday. Figure 2 shows the estimated number people are at the Mensa in each minute. On Wednesday, our results suggest the highest number of visitors at a time, with a maximal count of approximately 500 around 12:30, while Tuesday had the highest number of visitors overall. The predicted queues can be seen in Figure 3. The first plot compares the queue estimate of all the three models with their respective 95% confidence intervals. The computed standard errors and CI's have low values that make them hard to spot visually. The second plot in the figure shows the serving speed change depending on queue size. Here we can see the serving speed reaching its maximum of 20 around 11:30 and staying constant until 12:10, thereafter adapting with the variation of the queue.

4. Discussion & Conclusion

We obtained an estimate of the count of people at the Shedhalle canteen in each day with a meaningful sampling resolution of 1 minute from a single tracking device. We argue that the steps taken to quantify uncertainty enable a reasonable estimation for the number of people in a public space based on identified unique MAC addresses. We were also able to obtain descriptive results that showcase the time intervals where most Shedhalle visitors have lunch. Our initial visual investigation at the canteen was able to further confirm these counts, as we observed that the Mensa was operating at its full capacity on Wednesday, and we know from (vtechnik, 2019) that the total capacity is in fact 532 seats. (We also checked for the presence of our own mobile phone's mac id in the collected data to check for its accuracy.)

Next, we proposed three models for queue forming patterns. Our models experience issues with the collected data on Tuesday and Friday, where the queue increases even with

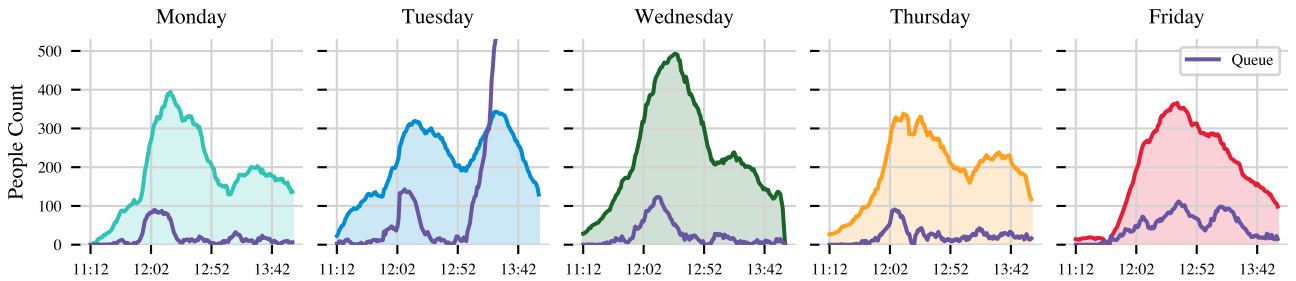


Figure 2. Histograms of the number of people and queue estimation using the 3rd model for each day of the week. The queue estimations resemble the shape of the total count of people, and it can be seen that the queue model explodes in Tuesday.

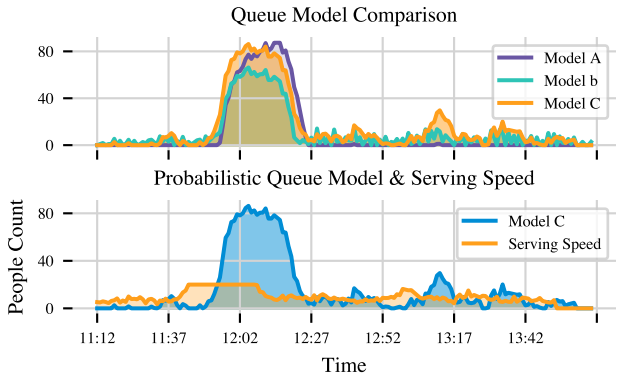


Figure 3. Upper row: Queue model estimates for the basline model (model A), informed deterministic (model B), and probabilistic model (model C) with 95% CIs for Monday. Second row: Illustration of serving speed influence on queue estimates.

decreasing total count. This stems from a device specific issue of recording many one-time mac addresses, that are first and last seen in the same minute. These spikes in mac addresses are not seen in the cumulative sum, since they cancel each other out, and only affect queue estimates.

Furthermore, when comparing the three models to each other, no significant difference can be seen in the descriptive data. We believe that in our case, estimating the queue from serving speed and first count was not a scenario that allowed for producing models with a lot of variance, as can also be seen in the confidence intervals of our models being so small. Still from the similar outputs of the models and our theoretical assumptions, we believe the estimates to be useful for someone that wants to avoid visiting the mensa at peak times.

Limitations and Future Research Directions

The main limiting factor for the generalizability of our estimates is the fact that we only tracked canteen attendance for one week. For evaluating if the variance between days is not caused by any other factors, further sampling is needed. An additional issue with the tracking is that we do not have a statistic that gives us an accurate estimation of how many students have their WiFi enabled on their phones. Regarding the queues, exact queue numbers are needed to validate the models.

Regarding tracking limitations, our method does not account for people who possess more than two devices. Furthermore, the device specific issues we encountered suggest that single-sensor tracking might not be fail-proof. For further investigations, identifying the underlying causes for higher queues at the canteen, like lecture scheduling and library visiting patterns naturally follow in order to understand queue patterns better.

Ethics and Legal Implications of WiFi Tracking

The European Data Protection Law (Parliament & of the European Union, 2016) imposes strict guidelines regarding consent, handling personal data and fair use in all situations involving public and private data. Due to the public nature of the data and the limitations in obtaining explicit consent from all visitors of the mensa without introducing any confounding effects, we resorted to the following measures to abide to these regulations. We did not attempt to decrypt any of the collected data and only used an aggregated version for our analysis, that makes the tracking or identification of a single device user impossible, therefore not putting the personal privacy of any student at risk. For grading, we only shared this report and a similar, more simplified dataset with the MOML group.

Contribution Statement

Paper presents a collaborative effort by a team where each member made significant contributions to various aspects of the project. All team members participated equally in data collection. Daniel was involved in data visualisation and project structuring. Sarthak focused on interpreting the data from device messages to device count. Philipp led data filtering and devised a method to transition from devices to people count. Mihai contributed to data interpretation and devised a method for computing the queue.

References

Android. Android randomization, 2024. URL https://source.android.com/docs/core/connect/wifi-mac-randomization-behavior?hl=de#:~:text=The%20MAC%20randomization%20feature%20randomizes,randomized%20MAC%20address%20by%20default. Website: source.android.com.

Apple. Apple randomization, 2021. URL https://support.apple.com/de-de/guide/security/secb9cb3140c/web#:~:text=In%20iOS%2014%20or%20later,in%20the%20Wi-Fi%20payload. Website: support.apple.com.

Fahrmeir, L., Heumann, C., Künstler, R., Pigeot, I., and Tutz, G. *Statistik: Der Weg zur Datenanalyse*. Springer-Verlag, 2016.

Martin, J., Mayberry, T., Donahue, C., Foppe, L., Brown, L., Riggins, C., Rye, E. C., and Brown, D. A study of MAC address randomization in mobile devices and when it fails. *CoRR*, abs/1703.02874, 2017. URL http://arxiv.org/abs/1703.02874.

Parliament, E. and of the European Union, C. Regulation (eu) 2016/679 of the european parliament and of the council of 27 april 2016 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data (general data protection regulation). https://eur-lex.europa.eu/eli/reg/2016/679/oj, April 2016. Accessed: [17.01.24].

Sapiezynski, P., Stopczynski, A., Gatej, R., and Lehmann, S. Tracking human mobility using wifi signals. *PloS one*, 10(7):e0130824, 2015.

Statista. Studierende in deutschland nach marke des eigenen handys oder smartphones im jahr 2020, 2020. URL https://de.statista.com/statistik/daten/studie/289103/umfrage/umfrage-unter-studierenden-zu-den-beliebtesten-handy-und-smartphonemarken/. Website: statista.com.

Traunmueller, M. W., Johnson, N., Malik, A., and Kontokosta, C. E. Digital footprints: Using wifi probe and locational data to analyze human mobility trajectories in cities. *Computers, Environment and Urban Systems*, 72: 4–12, 2018.

Tübingen. Bevölkerung von tübingen 2024, 2024. URL https://www.tuebingen.de/1370.html#:~:text=In%20T%C3%BCbingen%20leben%20derzeit%2091.249,bei%20etwa%2018%2C1%20Prozent. Website: tuebingen.de.

vtechnik. Mensa tübingen – interimslösung, 2019. URL https://vtechnik.de/projekt/mensa-tu%CC%88bingen-interimsloesung/. Website: vtechnik.de.

Wang, Z., Hong, T., Piette, M. A., and Pritoni, M. Inferring occupant counts from wi-fi data in buildings through machine learning. *Building and Environment*, 158:281–294, 2019.

Weppner, J., Bischke, B., and Lukowicz, P. Monitoring crowd condition in public spaces by tracking mobile consumer devices with wifi interface. In *Proceedings of the 2016 ACM International Joint Conference on Pervasive and Ubiquitous Computing: Adjunct*, pp. 1363–1371, 2016.